Lecture 11. Markov Chain Monte Carlo

Markov chains, ergodic theorem, Metropolis–Hastings, Gibbs sampler

Contents

11.1 Markov chains **2**

11.2 The Metropolis–Hastings algorithm **3**

11.3 The Gibbs sampler **4**

11.4 Practical issues **5**

11.5 Selected exercises **5**

Markov chain Monte Carlo (MCMC) simulates π by running a Markov chain with limiting distribution π and averaging along the path; an ergodic theorem replaces the law of large numbers.

11.1 Markov chains

A discrete-time Markov chain on a countable state space S is a sequence $\{X_n\}$ with

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \ldots, X_0) = P(X_{n+1} = j \mid X_n = i) = P_{ij}.$$

We assume time homogeneity. The rows of the **transition matrix** $P = (P_{ij})$ sum to one. Examples: simple random walk on $\mathbb{Z}$ ($P_{i,i+1} = p$, $P_{i,i-1} = 1-p$); gambler's ruin on $\{0, \ldots, N\}$, absorbing at 0 and N.

Chapman–Kolmogorov. $P_{ij}^{(n)} = P(X_{m+n} = j \mid X_m = i)$ satisfies $P^{(m+n)} = P^{(m)}P^{(n)}$ (condition on X_m), so $P^{(n)} = P^n$, and $X_0 \sim \pi^{(0)}$ (row vector) gives $X_n \sim \pi^{(0)}P^n$.

Classification.

- $i \to j$ (j *reachable*) if $P_{ij}^{(n)} > 0$ for some $n \geq 0$; i, j *communicate* if $i \to j$ and $j \to i$. The chain is **irreducible** if there is one communication class. The random walk is irreducible; gambler's ruin has classes $\{0\}$, $\{1, \ldots, N-1\}$, $\{N\}$.
- The **period** of i is $d(i) = \gcd\{n \geq 1 : P_{ii}^{(n)} > 0\}$; the chain is **aperiodic** if $d(i) = 1$ for all i.
- With $\tau_{ii} = \min\{n \geq 1 : X_n = i\}$ given $X_0 = i$, i is *recurrent* if $P(\tau_{ii} < \infty) = 1$, else *transient*; recurrent i is **positive recurrent** if $\mathbb{E}\tau_{ii} < \infty$, else null recurrent. These are class properties.
- **Ergodic** means aperiodic and positive recurrent. On a finite space an irreducible chain is positive recurrent, so irreducible and aperiodic means ergodic.

Stationary distributions. π is **stationary** if $\pi P = \pi$, $\pi \geq 0$, $\pi\mathbf{1} = 1$. If $P_{ij}^{(n)} \to \pi_j$ for all i, π is stationary: $\pi_k = \lim_n (P^{n+1})_{ik} = \sum_j \lim_n (P^n)_{ij} P_{jk} = (\pi P)_k$ (finite S; in general by dominated convergence).

Theorem 11.1 (Limit theorems; stated without proof)

(a) An irreducible chain has a unique stationary distribution if and only if it is positive recurrent.

(b) If the chain is irreducible and ergodic, then $\lim_n P_{ij}^{(n)} = \pi_j$ exists independently of i, π is the unique stationary distribution, and $\pi_j = 1/\mathbb{E}\tau_{jj}$ is also the long-run fraction of time spent at j.

(c) If the chain is irreducible and aperiodic and a stationary distribution π exists, then $P_{ij}^{(n)} \to \pi_j$, and the chain is positive recurrent, hence ergodic.

Theorem 11.2 (Ergodic theorem)

If $\{X_n\}$ is irreducible and ergodic with stationary distribution π, and $\mathbb{E}_\pi|g(X)| < \infty$, then

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to \mathbb{E}_\pi g(X) \quad \text{almost surely, from any starting state.}$$

This extends the strong law to Markov chains and underlies MCMC: build an irreducible, aperiodic, positive recurrent chain with stationary distribution f, run it, and average.

Detailed balance. Stationarity is usually guaranteed by **detailed balance**: $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j.

Proposition 11.3

If π satisfies detailed balance, it is stationary. Moreover the stationary chain is *reversible*: $(X_0, \dots, X_n)$ and $(X_n, \dots, X_0)$ have the same distribution.

Proof. $(\pi P)_j = \sum_i \pi_i P_{ij} = \pi_j \sum_i P_{ji} = \pi_j$. Reversibility: apply detailed balance repeatedly,

$$\pi_{i_0} P_{i_0 i_1} P_{i_1 i_2} \cdots P_{i_{n-1} i_n} = P_{i_1 i_0} \pi_{i_1} P_{i_1 i_2} \cdots = \cdots = \pi_{i_n} P_{i_n i_{n-1}} \cdots P_{i_1 i_0}.$$

■

General state spaces. P_{ij} becomes a **transition kernel** $K(x, y)$, the conditional density of X_{n+1} at y given $X_n = x$, $\int K(x, y)\, dy = 1$ (point masses allowed). Stationarity reads $\pi(y) = \int \pi(x) K(x, y)\, dx$, implied by $\pi(x)K(x, y) = \pi(y)K(y, x)$; the limit theorems carry over.

11.2 The Metropolis–Hastings algorithm

Motivation. Metropolis–Hastings builds a chain with the target as stationary distribution from density ratios alone, so normalizing constants are never needed.

Let f be the target (possibly unnormalized) and $g(\cdot \mid x)$ an easy **proposal**. Start at $X^{(0)}$ with $f(X^{(0)}) > 0$; for $t = 0, 1, \dots$:

(1) draw a candidate $\tilde{X} \sim g(\cdot \mid X^{(t)})$;

(2) compute the **Metropolis–Hastings ratio**

$$R(X^{(t)}, \tilde{X}) = \frac{f(\tilde{X})\, g(X^{(t)} \mid \tilde{X})}{f(X^{(t)})\, g(\tilde{X} \mid X^{(t)})};$$

(3) set $X^{(t+1)} = \tilde{X}$ with probability $\alpha(X^{(t)}, \tilde{X}) = \min\{R, 1\}$, and $X^{(t+1)} = X^{(t)}$ otherwise.

Metropolis et al. (1953) used a symmetric proposal $g(y \mid x) = g(x \mid y)$ (e.g. random walk $y = x + \epsilon$), giving $R = f(\tilde{X})/f(X^{(t)})$: uphill moves always accepted, downhill sometimes. Hastings (1970) allowed asymmetric g.

Theorem 11.4

The Metropolis–Hastings chain satisfies detailed balance with respect to f; hence f is a stationary distribution.

Proof. The transition kernel is

$$K(x, y) = \alpha(x, y) g(y \mid x) + \{1 - r(x)\}\delta_x(y), \qquad r(x) = \int \alpha(x, y) g(y \mid x)\, dy,$$

with $r(x)$ the acceptance probability from x. For the continuous part,

$$f(x)\alpha(x, y)g(y \mid x) = \min\{f(x)g(y \mid x),\ f(y)g(x \mid y)\} = f(y)\alpha(y, x)g(x \mid y),$$

the middle being symmetric in (x, y). The point mass $f(x)\{1 - r(x)\}\delta_x(y)$ lives at $y = x$, where it equals $f(y)\{1 - r(y)\}\delta_y(x)$. So $f(x)K(x, y) = f(y)K(y, x)$; apply Proposition 11.3. ■

The other conditions of Theorem 11.2 depend on g:

- *Irreducibility*: if $f > 0$ on S and the chain with kernel g is irreducible.
- *Aperiodicity* usually comes from rejections: e.g. in a discrete chain with $g(i \mid i) > 0$ and a proposable j with $f(j) < f(i)$, $\alpha(i,j) < 1$, so $r(i) < 1$, $K(i,i) > 0$ and $d(i) = 1$.

Example 11.1 (Posterior sampling)

For a posterior $p(\theta \mid y) = c\,p(y \mid \theta)p(\theta)$ with unknown c,

$$R(\theta, \tilde{\theta}) = \frac{p(y \mid \tilde{\theta})p(\tilde{\theta})g(\theta \mid \tilde{\theta})}{p(y \mid \theta)p(\theta)g(\tilde{\theta} \mid \theta)},$$

free of c. The **independence chain** $g(\cdot \mid \theta) = p(\cdot)$ (prior proposals) gives the likelihood ratio $R = p(y \mid \tilde{\theta})/p(y \mid \theta)$.
E.g. for $X \sim \text{Bin}(n,p)$, $p = e^\theta/(1+e^\theta)$, $\theta \sim N(0,\sigma^2)$ (Example 8.2), the posterior $\propto e^{\theta x}(1+e^\theta)^{-n}e^{-\theta^2/(2\sigma^2)}$ has no closed-form normalizer; propose $\tilde{\theta} \sim N(0,\sigma^2)$ and accept with probability $\min\{1, e^{(\tilde{\theta}-\theta)x}\{(1+e^\theta)/(1+e^{\tilde{\theta}})\}^n\}$.

11.3 The Gibbs sampler

For $X = (X_1, \ldots, X_p)$, with X_{-i} all coordinates but i, the Gibbs sampler updates coordinates in turn from their **full conditionals** (often standard):

$$X_i^{(t+1)} \sim f_{X_i \mid X_{-i}}(\cdot \mid X_1^{(t+1)}, \ldots, X_{i-1}^{(t+1)}, X_{i+1}^{(t)}, \ldots, X_p^{(t)}), \qquad i = 1, \ldots, p,$$

always using the latest values (as in Gauss–Seidel, Lecture 5).

Example 11.2 (Bivariate normal)

For $(X_1, X_2) \sim N(0, \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right))$, $X_1 \mid X_2 = x_2 \sim N(\rho x_2, 1-\rho^2)$ and symmetrically: $X_1^{(t+1)} \sim N(\rho X_2^{(t)}, 1-\rho^2)$, then $X_2^{(t+1)} \sim N(\rho X_1^{(t+1)}, 1-\rho^2)$. $X_2^{(t)}$ is autoregressive with coefficient ρ^2, so mixing is slow for $|\rho| \approx 1$.

Proposition 11.5

Each coordinate update of the Gibbs sampler is a Metropolis–Hastings step whose proposal is always accepted. Hence f is stationary for the Gibbs sampler.

Proof. Fix i and x. The proposal is $g_i(\tilde{x} \mid x) = f_{X_i \mid X_{-i}}(\tilde{x}_i \mid x_{-i})\,\delta(\tilde{x}_{-i} - x_{-i})$. On $\{\tilde{x}_{-i} = x_{-i}\}$ (probability one), factor $f(x) = f_{X_{-i}}(x_{-i}) f_{X_i \mid X_{-i}}(x_i \mid x_{-i})$:

$$R(x, \tilde{x}) = \frac{f_{X_{-i}}(x_{-i}) f_{X_i \mid X_{-i}}(\tilde{x}_i \mid x_{-i}) \cdot f_{X_i \mid X_{-i}}(x_i \mid x_{-i})}{f_{X_{-i}}(x_{-i}) f_{X_i \mid X_{-i}}(x_i \mid x_{-i}) \cdot f_{X_i \mid X_{-i}}(\tilde{x}_i \mid x_{-i})} = 1.$$

By Theorem 11.4 each update leaves f invariant, hence so does their composition. ■

So Gibbs is Metropolis–Hastings with a coordinate-dependent proposal (the composed chain need not be reversible, but f is stationary).

Example 11.3 (Ising model)

Place p spins $x_i \in \{-1, +1\}$ around a circle ($x_{p+1} = x_1$) with

$$p(x) = \frac{1}{Z} \exp\Big(\beta \sum_{j=1}^{p} x_j x_{j+1}\Big), \qquad \beta > 0,$$

favouring aligned neighbours. Z has 2^p terms, but the full conditional of x_i involves only its neighbours and Z cancels. With $s_i = x_{i-1} + x_{i+1}$,

$$P(x_i = +1 \mid x_{-i}) = \frac{e^{\beta s_i}}{e^{\beta s_i} + e^{-\beta s_i}} = \frac{1}{1 + e^{-2\beta s_i}},$$

logistic in the neighbours; equivalently the spin flips with probability $1/\{1 + \exp(2\beta x_i s_i)\}$.

11.4 Practical issues

- *Burn-in.* Discard the initial, non-stationary part of the run, judged by trace plots and convergence diagnostics.
- *Mixing and stopping.* Strong autocorrelation (Example 11.2 with $|\rho| \approx 1$, or a badly scaled random walk) makes the effective sample size much smaller than the number of iterations; Monte Carlo standard errors must account for it (e.g. batch means).
- *Dependence.* For nearly independent draws, run N parallel chains from dispersed starts and keep the final values, or *thin* one chain to $Y_m = X_{mk}$. Thinning saves storage, not error: for $\mathbb{E}_\pi g$ the full chain is never worse.

11.5 Selected exercises

Motivation. The two-state chain in closed form, slow mixing of the bivariate normal Gibbs sampler, and why thinning never helps.

Exercise 11.1 (The two-state chain)

Let $P = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$ with $a, b \in (0, 1]$, and put $\lambda = 1 - a - b$.
(a) Show that $\pi = \frac{1}{a+b}(b, a)$ satisfies detailed balance, hence is stationary.
(b) Show that $P^n = \Pi + \lambda^n E$ for $n \geq 1$, where $\Pi = \mathbf{1}\pi$ (both rows equal to π) and $E = \frac{1}{a+b}\left(\begin{smallmatrix} a & -a \\ -b & b \end{smallmatrix}\right)$.
(c) Deduce that $P^{(n)}_{ij} \to \pi_j$ geometrically fast unless $a = b = 1$, and explain the exception.

Solution. [Supplement]

(a) $\pi_1 P_{12} = \frac{ba}{a+b} = \pi_2 P_{21}$; apply Proposition 11.3.

(b)

Step 1. ($n = 1$.) $\Pi + \lambda E$ has $(1,1)$ entry $\frac{b+(1-a-b)a}{a+b} = 1 - a$, $(1,2)$ entry $\frac{a-(1-a-b)a}{a+b} = a$, and likewise b, $1 - b$ in row 2: $P = \Pi + \lambda E$.

Step 2. (Products.) $\Pi^2 = \mathbf{1}(\pi\mathbf{1})\pi = \Pi$; $E\mathbf{1} = 0$ gives $E\Pi = 0$; $\pi E = \frac{1}{(a+b)^2}(ba - ab, -ba + ab) = 0$ gives $\Pi E = 0$; and $E^2 = \frac{1}{(a+b)^2}\left(\begin{smallmatrix} a^2+ab & -a^2-ab \\ -ab-b^2 & ab+b^2 \end{smallmatrix}\right) = E$.

Step 3. (Induction.) If $P^n = \Pi + \lambda^n E$, then $P^{n+1} = (\Pi + \lambda^n E)(\Pi + \lambda E) = \Pi + \lambda^{n+1} E$ by Step 2.

(c) By (b), $|P^{(n)}_{ij} - \pi_j| \le |\lambda|^n \max\{a,b\}/(a+b)$, with $\lambda \in [-1,1)$ and $\lambda = -1$ only for $a = b = 1$; otherwise convergence is geometric at rate $|\lambda|$. For $a = b = 1$ the chain alternates deterministically (period 2), P^n alternates between I and P, and there is no limit, though $\pi = (\frac{1}{2}, \frac{1}{2})$ is stationary. ■

Exercise 11.2 (Mixing of the bivariate normal Gibbs sampler)

In Example 11.2, with $|\rho| < 1$:
(a) Show that $X_2^{(t+1)} = \rho^2 X_2^{(t)} + \varepsilon_t$ with $\varepsilon_t \sim N(0, 1-\rho^4)$ independent of $X_2^{(0)}, \dots, X_2^{(t)}$.
(b) Show that $N(0,1)$ is stationary for $(X_2^{(t)})$, and that started from it, $\operatorname{Corr}(X_2^{(t)}, X_2^{(t+k)}) = \rho^{2k}$.
(c) Show that for the stationary chain, $n \operatorname{Var}(\frac{1}{n}\sum_{t=1}^n X_2^{(t)}) \to \frac{1+\rho^2}{1-\rho^2}$, and interpret.

Solution. [Supplement]

(a) $X_1^{(t+1)} = \rho X_2^{(t)} + \sqrt{1-\rho^2} Z_1$, $X_2^{(t+1)} = \rho X_1^{(t+1)} + \sqrt{1-\rho^2} Z_2$, with $Z_1, Z_2 \sim N(0,1)$ independent of each other and the past, so

$$X_2^{(t+1)} = \rho^2 X_2^{(t)} + \varepsilon_t, \qquad \varepsilon_t = \rho\sqrt{1-\rho^2} Z_1 + \sqrt{1-\rho^2} Z_2,$$

and $\operatorname{Var} \varepsilon_t = (1-\rho^2)(\rho^2+1) = 1-\rho^4$.

(b) $X_2^{(t)} \sim N(0,1)$ gives $X_2^{(t+1)} \sim N(0, \rho^4 + 1 - \rho^4)$. Iterating (a), $X_2^{(t+k)} = \rho^{2k} X_2^{(t)}$ plus a combination of $\varepsilon_t, \dots, \varepsilon_{t+k-1}$, independent of $X_2^{(t)}$, so the correlation is ρ^{2k}.

(c) With $r = \rho^2 \in [0,1)$ and unit variances,

$$n \operatorname{Var}\Big(\frac{1}{n}\sum_{t=1}^n X_2^{(t)}\Big) = \frac{1}{n}\sum_{s,t=1}^n r^{|s-t|} = 1 + 2\sum_{k=1}^{n-1}\Big(1 - \frac{k}{n}\Big) r^k.$$

By monotone convergence this tends to $1 + 2r/(1-r) = (1+r)/(1-r)$: n Gibbs draws are worth about $n(1-\rho^2)/(1+\rho^2)$ independent ones (effective sample size), about $n/100$ for $\rho = 0.99$. ■

Exercise 11.3 (Thinning never helps)

Let (X_t) be a stationary sequence (the joint law of $(X_{t+s})_{t \ge 1}$ does not depend on s) and $\mathbb{E} g(X_1)^2 < \infty$. For integers $n, k \ge 1$ compare the full average and the thinned average

$$\bar g_{\text{full}} = \frac{1}{nk}\sum_{t=1}^{nk} g(X_t), \qquad \bar g_{\text{thin}} = \frac{1}{n}\sum_{m=1}^{n} g(X_{mk}).$$

Show that $\operatorname{Var} \bar g_{\text{full}} \le \operatorname{Var} \bar g_{\text{thin}}$.

Solution. [Supplement] For $j = 0, \dots, k-1$ let $A_j = \frac{1}{n}\sum_{m=1}^n g(X_{mk-j})$. The indices $mk - j$ cover $1, \dots, nk$ once, so $\bar g_{\text{full}} = \frac{1}{k}\sum_j A_j$. A_j is the same function of (X_{t-j}) as $A_0 = \bar g_{\text{thin}}$ of (X_t), so $\operatorname{Var} A_j = v := \operatorname{Var} \bar g_{\text{thin}}$ by stationarity, and $\operatorname{Cov}(A_i, A_j) \le v$ by Cauchy–Schwarz:

$$\operatorname{Var} \bar g_{\text{full}} = \frac{1}{k^2}\sum_{i,j=0}^{k-1} \operatorname{Cov}(A_i, A_j) \le \frac{1}{k^2} \cdot k^2 v = v.$$

Thinning is justified only by storage or the cost of evaluating g. ■